

DAEN/ISEN 427 Homework 3

Due date: Tue Sep 23, 2025 (before class)

Rubric: Full credit may be given for any reasonable Bayesian analysis.

1. Why is it that we identify ourselves at the very beginning of a phone conversation, even to people we already know, but we do not do this when we meet in person? Express your answer within the framework of Bayesian perceptual inference.

Answer:

2. When a conversation partner speaks softly, or when a conversation occurs in the presence of significant ambient noise, we sometimes cup our ears and/or look carefully at the speaker's lips. Why, in Bayesian perceptual terms, do we do this?

Answer:

3. Consider the following simplified version of the broken telephone game. A first player secretly communicates a message $m_1 \in \{0, 1, 2\}$ to a second player with probabilities

$$P(m_1 = 0) = P(m_1 = 1) = P(m_1 = 2) = \frac{1}{3}.$$

The second player then secretly communicates a message $m_2 = m_1 + n$ to a third (final) player where

$$P(n = 0) = \frac{2}{3} \quad P(n = +1) = \frac{1}{3}$$

- a. What is the optimal Bayes classification of the message received by the last player?

Answer:

- b. What is the expected error probability?

Answer: